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Competing discursive constructions of China's smog in Chinese and Anglo-American English-language newspapers: A corpus-assisted discourse study

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Abstract

This article presents a corpus-assisted discourse study of the representations of China's smog in one Chinese (i.e. *China Daily*) and three Anglo-American (i.e. *The New York Times*, *The Times* and *The Guardian*) English-language newspapers from 2011 to 2014. The findings suggest that they converge in representing China's smog as a kind of severe air pollution that has some consequences on residents in China and poses a problem that the government must tackle. However, the Chinese English-language newspaper prefers to represent it as a kind of weather phenomenon without serious impact on public health and to construct a positive and responsible image of the Chinese central government. The Anglo-American English-language newspapers are inclined to dramatize it as a disaster with a huge health impact, and construct a negative image of the Chinese government with a view to pressurizing it to take responsibility in the context of climate change.

Keywords

Air pollution, corpus-assisted discourse study, critical discourse analysis, environmental communication, media discourse, smog

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Introduction

As one of the most heavily populated areas in the world, China set its mind on economic development in the last three decades and transformed itself into one of the fastest growing economies in the world. This unprecedented economic boom, however, has begun to take its toll on China's already fragile environment in recent years, as can be witnessed from its worsening environmental problems such as air pollution, water pollution, sand storms and forest deterioration (Geall, 2013; Shapiro, 2012; Yao, 2008; Zhang and Barr, 2013). The enormity of China's environmental challenges has culminated in the breakout of the all-enveloping smog in large areas of China, especially in its capital Beijing, in 2013, which thrust China's environmental problems into the limelight of both domestic and international media (Chen, 2014b; Van de Ven, 2014).

It has been widely acknowledged that news media play a crucial and indispensable role in contemporary environmental issues (Hajer, 1995; Lester, 2010). It is especially influential in shaping public perception of issues about which they have no prior knowledge. However, news discourse is not a mere reflection of environmental problems that occur in daily life, but rather is socially produced, manufactured or constructed (Hansen, 1991, 2010). This social constructivist view of environmental news suggests the importance of investigating the representations of a given environmental issue by news media from different origins in view of the fact that different or competing ways of conceptualizing environmental issues may have significant implications for decisions and actions (Fløttum and Dahl, 2012).

Previous studies have exposed some significant findings as regards the representations of pollution in developed Western countries (Griffin and Dunwoody, 1997; Kenix, 2005; Ma, 2015; Major and Atwood, 2004; Mayer, 2012; Watson, 2014). For example, in a study on the coverage of air pollution in the US press, Kenix (2005) found that they failed to fulfill their social responsibility due to journalistic norms and the importance of advertising revenue. Mayer (2012) demonstrated that American broadsheets failed to make a direct link between asthma and air pollution. Major and Atwood (2004) also revealed that American newspapers tended to feature environmental stories without proposing the possible solutions.

Environmental communication in China has also been under extensive scrutiny in the past decade in view of its mounting environmental challenges (Day, 2005; Geall, 2011; Ma, 2015; Shapiro, 2012; Tilt, 2006; Tilt and Xiao, 2010; Tong, 2014; Xu, 2012; Yao, 2008). As Tong (2014) suggests, the construction of China's environmental problems in Chinese media and journalism is 'precarious', because it is subject to the interference of the state and the market. Nevertheless, the topic of environmental problems is not a complete taboo for journalism in China, because the central government welcomes media coverage of environmental problems to a certain degree in order to show its determination and capacities to tackle these problems (Tong, 2014). In addition, competing or contradictory voices can also be identified in Chinese media's representations of environmental problems (Tilt and Xiao, 2010; Tong, 2014). These studies have shed light on the uniqueness of environmental communication in China's media system.

As one of China's most urgent and thorny environmental challenges, smog and its communication concern not only China's social stability, but also its international image.

While its representations in Chinese media have aroused the interest of a few researchers (Chen, 2014a, 2014b; Ma, 2015; Xu, 2012), none has carried out a comparative study of the issue in different socio-political contexts. Apart from pointing out the 'substantial contradictions' (Xu, 2012) and 'important attitudinal shifts' (Chen, 2014b) in Chinese media's representations of the issue, they offer little information as regards to what extent the representations of the issue converge or diverge from the representations of environmental issues in dominant Western media, to say nothing of the complicated socio-political factors involved in their representations.

However, as Partington (2015) has suggested, a comparative perspective is necessary for discourse analysis, because it can reveal some discursive features that might otherwise be ignored. In view of this, the present study aims to contribute to the growing literature of China's environmental communication studies through a comparative study of the representations of China's smog in Chinese and Anglo-American English-language newspapers. It has two specific objectives: (1) to examine in what way and to what extent they converge and diverge in their representations of the issue, and (2) to explicate the socio-political factors behind their particular ways of representations.

Smog in China

The rise of smog as a national concern is a fairly recent phenomenon, even though air pollution, as a social problem, is no longer a fresh topic in China. It was not until 2008 that smog became a serious concern in China's public discourse, when the so-called 'Blue Sky' project was conducted in Beijing during the 2008 Summer Olympics. Nevertheless, it was the conflict between the Beijing municipal government and the local US embassy which awakened common Chinese people to the real dangers of smog. Before 2010, the majority of Chinese people had no prior knowledge about the so-called pollutant PM 2.5, that is, particulate matter of less than 2.5 μm in diameter, because the measurement of PM 2.5 levels was not included in China's air quality index. However, since 2008 the US embassy has monitored PM 2.5 levels in Beijing and published them on Twitter, provoking the first complaint from the Chinese government in 2009, warning that disclosing that information would incur unexpected social consequences. Later, the Chinese government publicly condemned the act as an infringement of China's national sovereignty. While the conflict came to nothing, it unexpectedly threw PM 2.5 into the media spotlight, awakening the Chinese public to the existence of such pollutants in the air, which they used to mistake for mere smoke and dust. The index provided by China's most powerful online search engine Baidu also confirms that the term PM 2.5 did not appear on the Chinese Internet until the beginning of 2011 (Van de Ven, 2014). Since then a number of key advances have been made by the central and local governments in the monitoring of PM 2.5 and smog in the air.

While there was never such an environmental issue in China as influential as smog, it does not mean that environmental problems did not exist before or that Chinese people were not aware of these problems in their life. The majority of Chinese people tolerated them owing to years of propaganda from the Chinese government, claiming that environmental problems are the necessary price to pay for modernization (Zhang and Barr, 2013). The primary reason for smog growing into a national concern might lie in the

words of Jack Ma, the renowned business tycoon in China. He once joked in a public speech that he was glad to see smog in China as it ended the privileges of the privileged class in China, because nobody could escape from the all-enveloping smog. Unlike other environmental problems whose impact was confined to local residents, smog affects not only common people but also social elites. In fact, the latter have taken the lead in China's anti-smog campaign. This places pressure on the Chinese government to release information about smog and to take measures to fight against it in order to placate them. The rise of smog as a prominent social and environmental issue in China can thus be attributed to the mutual influences of external and internal pressures. Therefore, the question that confronts the Chinese media is how to represent China's smog without impairing the image of the Chinese government in view of their ingrained political duties to the Chinese government.

Data and methodology

Data

The present analysis is based on two large corpora: the CD corpus and the AA corpus. The CD corpus collects all news texts in *China Daily* (CD), the largest national English-language newspaper in China, and the AA corpus consists of news texts from the three most influential quality English-language newspapers in the Anglo-American media system: *The New York Times* (NYT) from the United States and *The Times* (TT) and *The Guardian* (TG) from Britain (Hallin and Mancini, 2004, 2012). CD is selected because it serves the primary function of communicating the stances of the Chinese government to the world (Liu, 2015, 2017), featuring a reporting style which is considered most similar to the Anglo-American reporting style among Chinese newspapers (Mody, 2010). The three Anglo-American newspapers are chosen for two reasons. First, belonging to the same liberal media system, they are known for their professionalism in news reporting and their wide international influence. In addition, all of them gave an extensive coverage of China's smog (see Table 1). The time span set for data collection is from 2011 to 2014. The year 2011 was when the noxious air began to become a topic of day-to-day online and offline conversation in China (Van de Ven, 2014), and 2014 was the year when smog was at its worst (Chen, 2014b). All news texts mentioning the keywords *smog* and *China* were extracted from the electronic database LexisNexis and built into their respective corpora after a manual elimination of some irrelevant information in the news texts, such as copyright, load date and byline.

Table 1. General information about the two corpora.

	China Daily	Anglo-American news media		
		The Guardian	*The Times*	*The New York Times*
Number of texts	920	123	104	263
Tokens	506,189	117,224	87,844	242,213
Total	506,189	447,281		

Methodology

A corpus-assisted discourse study (CADS) approach is adopted for this study. The application of corpus linguistic methods to environmental communication studies has become increasingly popular in recent years, as can be witnessed in the growing literature in this area (Alexander, 2010; Bevitori, 2010, 2015; Grundmann and Krishnamurthy, 2010; Koteyko, 2010; Nerlich et al., 2012). However, most of these studies, as Baker et al. (2008) argue, fail to present a 'synergy' of corpus linguistics and (critical) discourse analysis, with only a few exceptions (Bevitori, 2010, 2015; Fløttum et al., 2014).

A CADS approach is known for its emphasis on the balanced combination of quantitative corpus linguistic methods with detailed qualitative discourse analysis (Baker et al., 2008; Partington, 2004, 2015). The present study combines corpus linguistic methods (such as keyword, concordance and collocate analyses) with critical discourse analysis (CDA). CDA emphasizes not only the close analysis of texts but also the explication of the social context of language use, and is keen on exposing the ideologies behind media discourse in view of the latter's role in the production, maintenance and change of social relations of power (Fairclough, 1995; Fowler, 1991; Van Dijk, 1998). Viewing ideology as 'the basis of the social representations shared by members of a group' (Van Dijk, 1998: 8), this study proposes to examine the ideologies behind their particular ways of representations.

With the corpus analytic tool Wordsmith 5.0, it starts from keyword analysis through comparing the two corpora, respectively, with a general corpus – the British National Corpus (BNC) 5 million sampler corpus – to identify the keywords of the two corpora. Keywords refer to words which are statistically more frequently used in one corpus when compared with their use in a secondary or general corpus (Baker, 2006). They can be ranked in a keyword list by their keyness values, which are based on the calculation of log-likelihood. The more statistically significant a keyword is, the higher its keyness value.

This study focuses on the top 100 lexical keywords in each corpus based on the assumption that they can reflect the two corpora's particular preferences for the representations of the issue (Baker et al., 2013). They are further examined in terms of their potential contributions to the particular ways of framing China's smog because news frames can be 'manifested by the presence and absence of certain keywords' (Entman, 1993: 52). The particular ways of framing an issue carry significant ideological meanings because they can encourage particular understandings of the issue through selecting and highlighting some aspects of it (Alexander, 2010; Haarman and Lombardo, 2009; Touri and Koteyko, 2015). Some of the top 100 lexical keywords (see Table 2) are further selected and categorized into the four functions of a frame (Entman, 1993), as in the following:

1. **Problem definition:** keywords concerning the nature of the problem;
2. **Cause:** keywords suggesting the causes of the problem;
3. **Consequence:** keywords pointing to the consequences of the problem; and
4. **Solution/action:** keywords indicating possible solutions/actions to the problem.

It has to be pointed out that the classification of keywords cannot be clear-cut, because some of the keywords may serve multiple functions in the same corpus. Take *residents*

Table 2. Selected keywords for further analysis in the two corpora.

	Problem definition	Cause	Consequence	Solution/action
CD and AA	*Smog, air, pollution, environmental, quality, haze, polluted, growth, particles, PM*	*Emissions, coal, vehicles, cars, traffic, carbon pollutants*	*Residents*	*Masks, purifiers, government, clean, energy*
CD	*Meteorological, skies, blue, weather, heavy, alert*	*Fireworks*	*Economic, development, tourism, marathon, tourists, inbound, APEC*	*Measures, reduce, efforts, reform*
AA	*Greenhouse, climate, hazardous*	*Factories, fuels, oil, fossil, dioxide, burning*	*Lung, health*	

APEC: Asia-Pacific Economic Cooperation.

in Table 2 for example; although it has also been occasionally used to construct residents as potential contributors to air pollution in CD, its frequent use in both corpora addresses primarily public concerns about the potential impact of smog on them, as can be witnessed in most of its occurrences in both corpora (see the analysis in the 'Consequence' sub-section later), so its primary function is to frame the consequences of air pollution on residents. Nevertheless, these identified keywords can at least suggest the preferred perspectives of representations and indicate the entry points for further analysis (Baker et al., 2008).

As shown in Table 2, some keywords can be identified in both corpora, whereas others can only be identified in a single corpus. A close examination of the contextual uses of these keywords is required to gain an accurate understanding of their exact roles in representing China's smog, because even the shared keywords may be employed to construct different or even competing frames of the issue. Therefore, these keywords are then selected for further concordance and/or collocate analysis. Collocates refer to the words which occur in the neighborhood (usually five words to the left and right as the default) of a search word, while a concordance refers to a list of all of the occurrences of a search word along with the context in which they occur (Baker, 2006). The whole study is characterized by the constant movement between these keywords and the detailed discourse analysis of them in their specific contexts of use in order to determine their contributions to the particular representations of China's smog in the two corpora (Morley and Bayley, 2009).

Findings

Problem definition

As shown in Table 2, the two corpora share many keywords referring to the smog problem, such as *smog*, *air*, *pollution*, *environmental*, *quality*, *haze*, *polluted*, *growth*, *particles* and *PM*. This suggests that both corpora represent smog as a kind of air pollution

Table 3. Top 10 evaluative lexical collocates at the L1 position of *smog* in CD and AA.

Rank	CD		AA	
	Collocates	Frequency	Collocates	Frequency
1	*Heavy*	367	*Choking*	19
2	*Worst*	74	*Heavy*	18
3	*Severe*	57	*Toxic*	14
4	*Thick*	57	*Thick*	13
5	*Anti*	51	*Hazardous*	7
6	*Choking*	34	*Severe*	6
7	*Lingering*	31	*Acrid*	5
8	*Serious*	15	*Killer*	5
9	*Notorious*	13	*Urban*	4
10	*Dense*	11	*Airpocalypse*	3

and an environmental problem, which usually appears as haze and is measured by the amount of particles of a particular size, especially PM 2.5, present in the air. However, CD and AA are noted for their preferences for two evaluative keywords, *heavy* and *hazardous*. A close examination of their collocates in the two corpora reveals that they are used primarily to describe smog. While *heavy* describes only the compositional nature of smog, *hazardous* emphasizes the impact of smog on people (Martin and White, 2008 [2005]). It suggests that the two corpora might have different ways of evaluating smog. Since these evaluative words tend to occur immediately before smog as a modifier, the top 10 evaluative lexical collocates in the L1 position (i.e. the first position to the left) of *smog* are further examined (see Table 3).

As shown in Table 3, CD is noted for its overwhelming use of *heavy*. Together with *thick* and *dense*, they only capture the compositional nature of smog. Although some other evaluative words (such as *worst*, *severe*, *serious* and *notorious*) also highlight the severity of smog, they only describe smog as an unexpected and bad natural phenomenon without impact on public health. They can be equally used to describe such natural phenomena as rain, snow and fog. In contrast, AA is characterized by its preference for these evaluative words which dramatize the poisonous and lethal nature of smog to human beings, such as *choking*, *toxic*, *hazardous*, *acrid*, *killer*, *airpocalypse* and *noxious*. These distinct ways of evaluation contribute to their preferential ways of framing the issue and shaping the public's understanding of it (Stibbe, 2015). While CD frames smog as a severe natural phenomenon, AA tends to frame it as a disaster or 'airpocalypse'.

Their preferential ways of framing the issue can also be reflected in CD's preferences for some keywords related to weather conditions (*meteorological*, *skies*, *blue* and *weather*) and AA's preferences for those keywords concerning climate (*greenhouse* and *climate*). In CD, the most frequent lexical collocates of *weather* (275, 0.05%) are *smoggy* (61), *conditions* (30), *said* (24), *smog* (17), *pollution* (17) and *forecast* (16). They indicate that CD represents smog primarily as a kind of weather condition. Take its strongest collocate *smoggy* as an example: 59 out of 61 occurrences collocate with *weather* in the L1 position, forming the nominal group *smoggy weather*, helping to conceptualize smog

as a specific property of weather. This particular way of conceptualization has significant ideological implications. First, through underlining the common and everyday nature of the problem, its seriousness is downplayed. In fact, the impact of this particular way of framing is quite deep, since many Chinese used to perceive smog in this way. In addition, it helps to disguise the causes of smog. As Chen (2014a) suggests, the representation of smog as a kind of weather condition suggests that 'the smog would soon dissipate, just like other bad weather *caused by non-human factors*' (p. 84). Since smog change is framed as weather change, it is not a cause for surprise or alarm. More importantly, it also presupposes the government's role in this issue. The government is only responsible for releasing information about smog or issuing smog alerts, just like releasing information about rain or storms.

In contrast, AA is noted for its preference for keywords concerning climate change. The most frequent lexical collocates of *climate* (175, 0.03%) in AA are *change* (259), *global* (48), *energy* (34), *China* (28) and *Paris* (26), suggesting that AA represents smog primarily in terms of climate change. In its representations of China's smog, AA constructs China as a main contributor to climate change, as in the following:

(1) Yet burning coal is also highly polluting, both in producing *smog* and greenhouse gases that have been linked by many scientists to *climate change*. (NYT, 17/11/2013)

(2) China's coal plants and vehicles emit a suite of emissions, including sulfur dioxide, nitrogen oxides, carbon dioxide, mercury and small particles (soot). Some pollutants, including carbon dioxide, contribute to the worldwide problem of *climate change*. (NYT, 26/12/2013)

The framing of smog in terms of climate change constructs smog as a problem concerning not only China but also the world. Through representing China's smog, AA takes it as an opportunity to push the Chinese government towards reforming its environmental policies and fighting against climate change, as in the following:

(3) That led to discussions about how weaning Chinese industries off coal would not just clean the air, but would also permit China to make global commitments in the battle against *climate* change, the insiders said. (NYT, 13/11/2014)

Cause

Several keywords can be identified in the two corpora, that are supposed to be related to the representations of the causes of smog, such as *emissions*, *coal*, *vehicles*, *cars*, *traffic*, *carbon* and *pollutants*. This suggests that both corpora represent emissions and pollutants from coal burning and vehicles as the main causes for smog. This is consistent with previous findings of air pollution studies (Bickerstaff and Walker, 2001; Gonzalez, 2005). Air pollution has often been attributed to 'rising urban populations, unchecked urban and industrial expansion and the phenomenal surge in the number and use of motor vehicles' (Bickerstaff and Walker, 2001: 133).

However, the two corpora differ significantly in their ways of representing causes. Take *emissions* as an example: the most frequent lexical collocates of *emissions* (524, 0.10%) in CD include *cut* (75), *carbon* (62), *reduce* (59), *energy* (48) and *dioxide* (45).

This suggests that CD emphasizes what central and local governments have done or will do to deal with carbon emissions, as in the following:

(4) Ding estimated the new standard will cut nitrogen oxide *emissions* by 300,000 metric tons annually. (CD, 21/12/2013)

(5) She said that when Tianjin issues a yellow alert, the local government requires about 100 key businesses to cut *emissions* by 20 percent. (CD, 12/03/2014)

Nevertheless, in AA, the most frequent collocates of *emissions* (577, 0.13%) are *carbon* (135), *peak* (65), *reduce* (59), *dioxide* (49) and *China's* (42). *Emissions* frequently collocates with *peak* because AA is concerned about when China's carbon emissions will peak, as in the following:

(6) He points out that for years researchers have projected that China's CO2 emissions would *peak* around 2030. (TG, 12/11/2014)

(7) To surprise and applause, China pledged that its emissions would *peak* by around 2030 and that, by the same year, alternative energy would provide around 20 percent of its needs. (NYT, 16/12/2014)

Their different ways of representing emissions actually reflect the two corpora's different ideologies in addressing emissions. While CD emphasizes the short-term or temporary aim of reducing emissions, AA is keen on pushing the Chinese government to set the ultimate aim in emission control.

However, CD is also noted for its preference for the keyword *fireworks*. A close examination of its contextual uses suggests that CD tends to highlight fireworks as one of the main causes for smog (see Example 8). It has 358 (0.07%) occurrences in CD, but only 74 (0.02%) occurrences in AA.

(8) However, with regular smog the contribution of *fireworks* to air pollution has drawn the attention of authorities. (CD, 25/12/2014)

It is a tradition for Chinese people to set off fireworks to celebrate important festivals, especially the Spring Festival. The emphasis on fireworks contributes to the shift of the responsibilities for China's smog to common Chinese people. This has resulted in firework bans in many cities in China. Nevertheless, smog conditions continue to get worse, so the representations of fireworks as one of the main causes for smog are quite disputable. It can be interpreted as an attempt to ameliorate more direct criticism of the Chinese government (Chen, 2014a).

However, AA is found to prefer such keywords as *factories*, *fuels*, *oil*, *fossil* and *dioxide*, suggesting that AA emphasizes the contribution of factories as well as the consumption of not just coal but also fossil fuels as the main causes for smog, as in the following:

(9) Coal-burning *factories* were the biggest sources of pollutants and greenhouse gases, which contribute to global warming. (NYT, 21/01/2014)

(10) Many *factories* sit in the Yangtze River Delta area of Jiangsu, and the area is one of the most polluted outside of northern China. (NYT, 05/11/2013)

While emphasizing factories as the main contributors to smog in China, AA also satirizes the Chinese government's diffusion of responsibility to common Chinese people while failing to address the real causes of smog, as in the following:

(11) With smoke billowing from the chimneys of *factories* in the distance, officials promoted the slogan: 'Curbing air pollution starts with me!' (NYT, 30/10/2013)

Consequence

In terms of the consequences of smog, both corpora are noted for their preferences for the keyword *residents*. Residents represent the key party that is likely to be affected by smog. The occurrences of *residents* are 321 (0.06%) in CD and 201 (0.04%) in AA. The most frequent collocates of *residents* in CD are *Beijing* (51), *urban* (38), *many* (30), *local* (27) and *air* (21); in AA, they are *Beijing* (38), *pollution* (16), *Chinese* (12), *cities* (10) and *local* (10). It can be seen that although both urban and local residents are mentioned, the two corpora prefer to focus on residents in urban areas. Since *Beijing* is always the most frequent collocate of *residents* in both corpora, the collocation of *Beijing* and *residents* is further examined with a view to exploring their ways of representing the consequences of smog for Beijing residents.

A close examination of the representations of residents in Beijing in CD has revealed that it constructs them as either contributors to smog (see Example 12) or beneficiaries of the Chinese or Beijing government's policies and actions (see Example 13):

(12) *Beijing residents* have been **asked** to replace fireworks with flowers and electronic substitutes. (CD, 01/02/2014)

(13) The reduction in the number of vehicles on the road by half, along with other temporary measures, **enabled** *Beijing residents* to **enjoy** blue skies during the just concluded Asia-Pacific Economic Cooperation meeting. (CD, 22/11/2014)

Example 12 implies that if Beijing residents stop using fireworks there will be less smog. It constructs Beijing residents as contributors to smog and the Beijing government as a responsible regulator. Example 13 directly constructs Beijing residents as the beneficiaries of the Beijing government's traffic control measures during the Asia-Pacific Economic Cooperation (APEC) meeting, ignoring the inconvenience such measures might bring to residents who need to get around the city.

There is only one case in which Beijing residents' uneasiness about smog is mentioned, but this negative reaction has been dismissed with the concessive *but*, as in the following:

(14) *Beijing residents* will no doubt feel uneasy when the city is once again smothered in smog after the APEC meeting, ***but*** they will get used to it as they did before. (CD, 05/11/2014)

The concessive *but* is known for its contrastive and concessive capacity (Dahl and Fløttum, 2014). While acknowledging the stance pointing at the potential uneasiness smog might bring about in Beijing residents, the reporter sets up a more important voice following *but*, which emphasizes that they will get used to it as they did before.

By contrast, AA shows a different picture in its construction of residents in Beijing. The majority of clauses in which *residents* and *Beijing* co-occur represent residents in Beijing directly as the actors of the process, and focus on the latters' thoughts, complaints and sufferings, as in the following:

(15) *Residents* of *Beijing* **want** to know why the Chinese government's air pollution readings differ from those of the American Embassy. (NYT, 01/11/2011)

(16) This month, *residents* of *Beijing* have already **suffered** through several days of hazardous air. (NYT, 24/10/2013)

Their different emphases on the consequences of smog can also be perceived through their preferences for some other keywords. For example, CD is noted for its preference for *economic*, *development*, *tourism*, *marathon*, *tourists*, *inbound* and *APEC*. The emphasis on *economic* and *development* suggests that China is still concerned about the priority of economic development. This can also be witnessed in its preference for *tourism*, *tourists* and *inbound*, because the tourism industry, inbound tourism in particular, is most vulnerable to the influences of smog. While *marathon* and *APEC* represent the two important events held in Beijing during the period, that is, the Beijing Marathon and the APEC meetings held in Beijing in 2014, one interesting question is how CD represents China's priority for economic development in terms of its serious environmental impact, so a close examination of *economic* was conducted.

In CD, *economic* (787, 0.16%) collocates most frequently with *growth* (147) and *development* (128), indicating that CD is concerned primarily with economic growth and development. In order to examine how CD represents the influences of smog or environmental pollution on economic development and growth, the tokens of *economic development* and *economic growth* are further examined in their contexts of use. The total occurrences of *economic development* and *economic growth* are 142 and 87 respectively. Detailed analyses have revealed that the priority of economic development and growth has not been challenged (Tong, 2014). Instead, CD repeatedly emphasizes the compatibility of economic development and environmental protection as well as the necessity for sustainable economic growth and development, as in the following:

(17) Sheng Fulai, head of the Research and Partnerships Unit of UNEP in Geneva, says *economic development* and ***environmental protection*** are ***not*** mutually exclusive. (CD, 01/07/2014)

(18) *Economic development* is ***not*** necessarily accompanied by ***pollution***. But in the economic history of the mankind, economic activities very often lead to pollution and environmental degradation. It is happening in China now, and it happened in Western countries decades ago. (CD, 23/10/2014)

The use of polemic negation in the previous examples is certainly dialogic and refutative (Martin and White, 2008 [2005]). While justifying the priority of economic development, CD tries to reject the prevalent ideas that economic development is bound to produce pollution, and that economic development and environmental protection are mutually exclusive. This ideology can also be detected through other discursive strategies, as in the following:

(19) Since taking office, President Xi Jinping and Premier Li Keqiang have pledged many times to better protect the environment, reaffirming the importance of **balancing** *economic development* and ***environmental protection***. (CD, 05/03/2014)

(20) Chinese policymakers have also stressed fostering ***greener*** *economic growth* and pushing forward similar urban transitions. (CD, 11/03/2014)

The first example metaphorically constructs economic development and environmental protection as a machine through the word *balancing*. It implies that the dual purpose of economic development and environmental protection can be achieved through the enhancement of management. The second example moves a step further to metaphorically construct economic growth as a plant through the word *greener*, which implies that economic development can be inherently environmentally friendly. It is consistent with Xu's (2012) finding that media in emerging economies seldom question the priority for constant economic growth.

In contrast, AA tends to emphasize the impact of smog on health, as can be seen from the two keywords *lung* and *health* in AA. The token *lung* has 75 (0.02%) occurrences in AA. It collocates most frequently with *cancer* (44), with 39 instances occurring at the R1 position (i.e. the first position to the right) of *lung*. An examination of the collocation of *lung* and *cancer* has revealed that AA represents the rapid rise of lung cancer rates in China as one of the severe consequences of smog, as in the following:

(21) A type of *lung cancer* reported to be increasing in Beijing has been linked to worsening air quality, with an expert warning that the potential health impact could be much greater than the SARS epidemic in 2003. (CD, 27/02/2014)

The token *health* has 402 (0.09%) occurrences in AA. Its most frequent collocates include *world* (82), *organization* (40), *public* (37), *pollution* (27), *organization* (26), *air* (23) and *human* (21). In order to examine the representations of the impact of air pollution on health, the collocation of *pollution* and *health* was further examined:

(22) Air *pollution* is clearly linked to *health*, and the great thing is that everybody – that's government officials and company executives alike – breathes the same air. (NYT, 01/09/2013)

As Example 22 shows, AA tends to make a direct link between air pollution and smog, which is in contrast to Mayer's (2012) finding that American newspapers failed to make a direct link between air pollution and respiratory diseases like asthma.

Solution/action

The two corpora are found to prefer the use of five keywords: *masks*, *purifiers*, *government*, *clean* and *energy*. The keyword *energy* is related to the solution of the problem in the two newspapers because it occurs frequently with the keyword *clean* in both corpora, suggesting that they tend to value clean energy as a possible solution to air pollution. *Masks* and *purifiers* suggest what common people do to deal with smog, as can be witnessed in their most frequently used lexical collocates. *Masks* collocates frequently with lexical words such as *face* (43), *wearing* (34) and *wear* (33) in CD, and with the lexical words *face* (42), *air* (16), *purifiers* (12) and *wear* (12) in AA. It suggests that both corpora cover people's wearing of masks as a way to protect themselves from smog. *Purifiers* is used in both corpora to refer to air purifiers because it collocates overwhelmingly at the L1 position with *air* in both corpora.

However, the two corpora show distinct differences in their representations of masks. CD represents masks primarily in actional processes (Fowler, 2013), simply highlighting the fact that some people are wearing masks to protect themselves from smog. It neither highlights its necessity nor questions its effectiveness, as in the following:

(23) Tourists **wear** filter *masks* in Beijing on Wednesday to avoid inhaling pollutants. (CD, 26/12/2013)

(24) Some runners **wear** *masks* to protect themselves from heavy smog as they take part in the Beijing Marathon on Sunday. (CD, 20/10/2014)

This is in sharp contrast to AA's representations of masks. Rather than describing the action of wearing masks, AA is concerned about the quality of the masks and their effectiveness in preventing smog, as in the following:

(25) Many *masks* worn around Asia are **simple surgical-type masks**. (NYT, 20/03/2014)

(26) 'Consumers **simply just don't know** which of those *masks* are **the best**', Richard Saint Cyr, a physician in Beijing who writes a health column… (NYT, 20/03/2014)

The representations of *government* in the two corpora also show apparent differences. *Government* has 1005 (0.20%) occurrences in CD and 842 (0.19%) occurrences in AA. In CD, the most frequent lexical collocates of *government* are *central* (117), *municipal* (90), *said* (81), *Beijing* (60), *local* (48) and *Chinese* (48). In AA, they are *Chinese* (107), *said* (47), *Beijing* (34), *central* (33), *pollution* (27) and *China's* (23). They suggest that CD is inclined to make a distinction between central (*central*) and local governments (including *Beijing*, *local* and *municipal*), while AA tends to focus on the Chinese government as a whole. A close examination of the collocation of *Chinese* and *government* in AA suggests that AA tends to represent the Chinese government from a negative perspective by emphasizing the inaccuracy of the Chinese government's air quality index, as in the following:

(27) The Chinese *government* describes air with an AQI between 301 and 500 as '**heavily polluted**' and urges people to refrain from exercising outdoors; the elderly and other

> vulnerable populations are supposed to stay indoors entirely. The U.S. Environmental Protection Agency uses a similar index that labels any reading between 301 and 500 as '**hazardous**'. (NYT, 21/10/2013)

Although CD tends to portray the Chinese government's performance in environmental protection from a positive perspective, it tends to present divergent evaluations of the central and local governments. In its representations of the central government, it prefers to highlight its efforts, measures and contributions towards air pollution and smog control. This can also be witnessed in CD's preference for some other keywords, such as *measures*, *alert*, *reduce*, *efforts* and *reform* (see Table 2), as in the following:

(28) The central *government* unveils the Airborne Pollution Prevention and Control Action Plan (2013–17), the toughest measure to date to tackle the worsening air pollution. (CD, 16/01/2014)

Nevertheless, CD also delivers occasional criticisms towards local Chinese government's impotence and failures in environmental control, especially during periods of severe smog, as in the following:

(29) Heavy smog has continued for at least three days, and there has been **mounting criticism over the inaction** of the municipal *government*. (CD, 16/02/2014)

The divergent ways of representing the central and local Chinese governments in CD help to construct a positive image of the central government through shifting the blame for air pollution to local government, especially during periods of severe smog.

Conclusion

A comparative study of Chinese and Anglo-American English-language newspapers has revealed both similarities and differences in their representations of China's smog. First, while both corpora represent the issue as a kind of air pollution and environmental problem, CD prefers to represent it as a natural phenomenon without serious impact on public health, but AA tends to dramatize it as a disaster which contributes to climate change. In addition, although both corpora emphasize emissions from coal burning and vehicles as the main causes for the problem, CD also underlines fireworks as one of the main causes of smog or air pollution. As regards the consequences of smog, both corpora treat residents as the key party in their representations, but CD tends to downplay smog's impact on residents, while AA underlines residents' complaints and suffering. Moreover, CD is still concerned about the priority of economic development, while AA is concerned primarily about its impact on public health. Finally, while the two corpora feature the government's performance in environmental protection, CD tends to construct a positive image of the central government, sometimes at the expense of the local government. However, AA tends to construct a negative image of the Chinese government through emphasizing its inaction and artifice concerning the issue of smog control.

A proper understanding of their different ways of representing smog cannot be acquired without situating them in the socio-political contexts in which they are embedded. As

suggested by Tong (2014), the Chinese government still holds strong control over the coverage of environmental problems because it might cause damage to its image, but the severity of China's air pollution also requires the Chinese government to 'communicate to the public about its determination and ability to tackle these problems' (p. 349), so the communication of environmental problems is also welcomed to a certain degree. Therefore, while CD is quite open about the severity of China's smog, it still serves the primary function of constructing a positive image of the Chinese government, though in a more strategic way, thus featuring the dual purpose of political correctness and journalistic objectivity in its environmental communication (Geall, 2011).

In contrast, AA's extensive coverage of China's smog and accentuation on its harmful effects on health do not mean that they are concerned about the welfare of the Chinese public but rather serve important ideological functions, that is, to construct a negative image of the Chinese government and push it to reform its environmental policies and take responsibility for fighting against climate change. This can also be further perceived through AA's ambivalent stance in the representations of environmental pollution in their own countries (Fowler, 1991; Kenix, 2005; Watson, 2014). The dramatic representations of China's smog in Anglo-American media are thus unsurprising, because they are consistent with the 'anti-China' ideology that has long been cultivated in Anglo-American media (Stone and Xiao, 2007) and the anxious efforts of the British and the American governments to pressurize the Chinese government to cap its emissions and contribute to climate change control. The representations of an environmental issue in a foreign country in Anglo-American media are thus subject to the local prism of national interests (Lee et al., 2002). Therefore, a comparative corpus-assisted discourse study of the representations of the same issue by media from different socio-political contexts can contribute to the exposure of not only their particular ways of representing the issue, but also, more importantly, the influences of different socio-political factors lying behind their representations.

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